



Deep Reinforcement Learning for UAV Placement in 3D Urban Environments

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Outline

I. Introduction

II. System model

III. DRL model

IV. Result

V. Conclusion

Motivation

UAV (Unmanned Aerial Vehicle) is an aircraft that operates **without a human pilot onboard** and can be controlled remotely or autonomously.

Why UAV Networks?

- UAVs provide temporary aerial coverage.
- High traffic demand during large events.
- Flexible deployment and wide coverage.
- Smart placement improves network performance.

Why intelligent UAV placement?

- UAV coverage changes dynamically
- User locations continuously move
- Static placement is inefficient
- Need autonomous decision-making

Motivation of this work

- Use Reinforcement Learning for autonomous UAV positioning
- Improve urban user coverage using real city environment



Background of UAV Network and DRL

UAV Networks

- UAV acts as aerial base station
- Provides wireless coverage to ground users
- Flexible and fast deployment

Deep Reinforcement Learning (DRL)

- Learns through rewards and penalties
- No predefined movement rules required
- Suitable for dynamic environments

Goal

- Maximize user coverage
- Improve Line-of-Sight communication
- Learn intelligent UAV movement

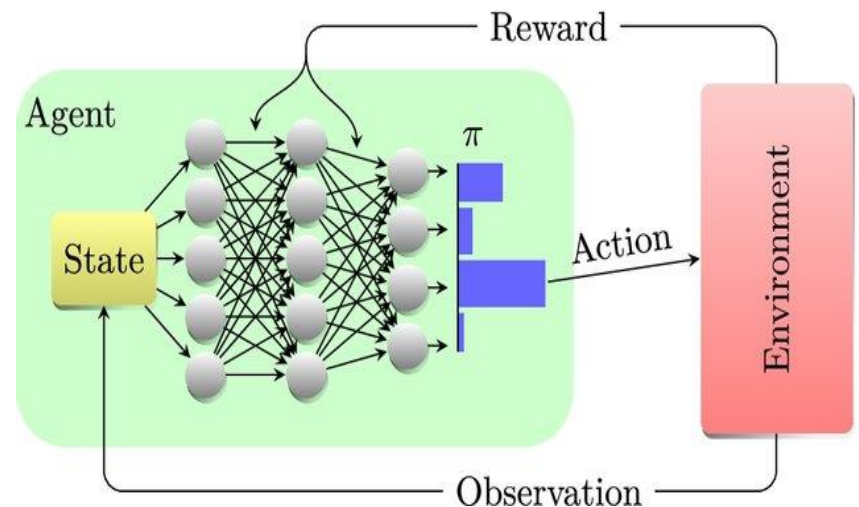


Fig: Deep reinforcement learning scheme.

Problem Statement

Existing Problems

- Static UAV placement is inefficient
- Urban buildings block communication
- Users move dynamically
- Coverage changes continuously

Challenges

- Realistic urban blockage
- Dynamic user mobility
- Efficient exploration
- Stable reinforcement learning

Objective

Develop an intelligent UAV placement system using Double DQN in Urban environment

Problem Formulation

Objective

Maximize coverage and communication quality while minimizing overlap between UAVs.

$$\max J = \text{Coverage} + \text{LoS} + \text{Density} + \text{Separation} - \text{Overlap}$$

Where:

- **Coverage** = Number of users served
- **LoS** = Line-of-Sight communication quality
- **Density** = Moving toward dense user clusters
- **Separation** = Keeping UAVs sufficiently apart
- **Overlap** = Redundant coverage between UAVs

Distance Between UAV and User

$$d_i = \sqrt{(x_u - x_i)^2 + (y_u - y_i)^2}$$

where:

- (x_u, y_u) = UAV position
- (x_i, y_i) = User position

Reward Function

$$R = w_1C + w_2D + w_3L + w_4S - w_5O$$

Where:

- C = Coverage Reward
- D = Density Reward
- L = LoS Reward
- S = Separation Reward
- O = Overlap Penalty

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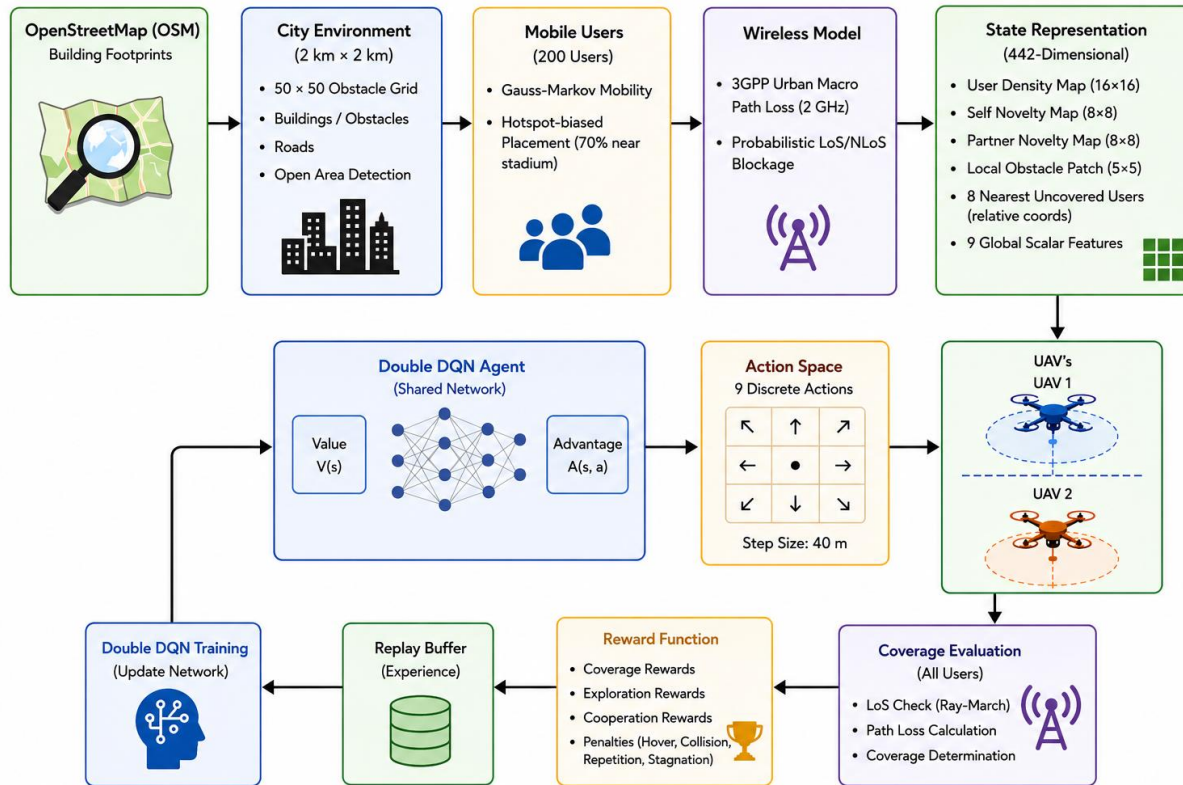
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Overall System Architecture

System Architecture



Components

- Real City Map (OpenStreetMap)
- UAV Agent
- Mobile Users
- Urban Buildings
- Reinforcement Learning Environment
- Double DQN Agent

Workflow

- UAV observes environment
- Selects movement action
- Receives reward
- Learns optimal coverage behaviour

Real City Environment Creation

Map Source

OpenStreetMap (OSM)

Tools Used

- OSMnx
- GeoPandas
- Shapely

Selected Area

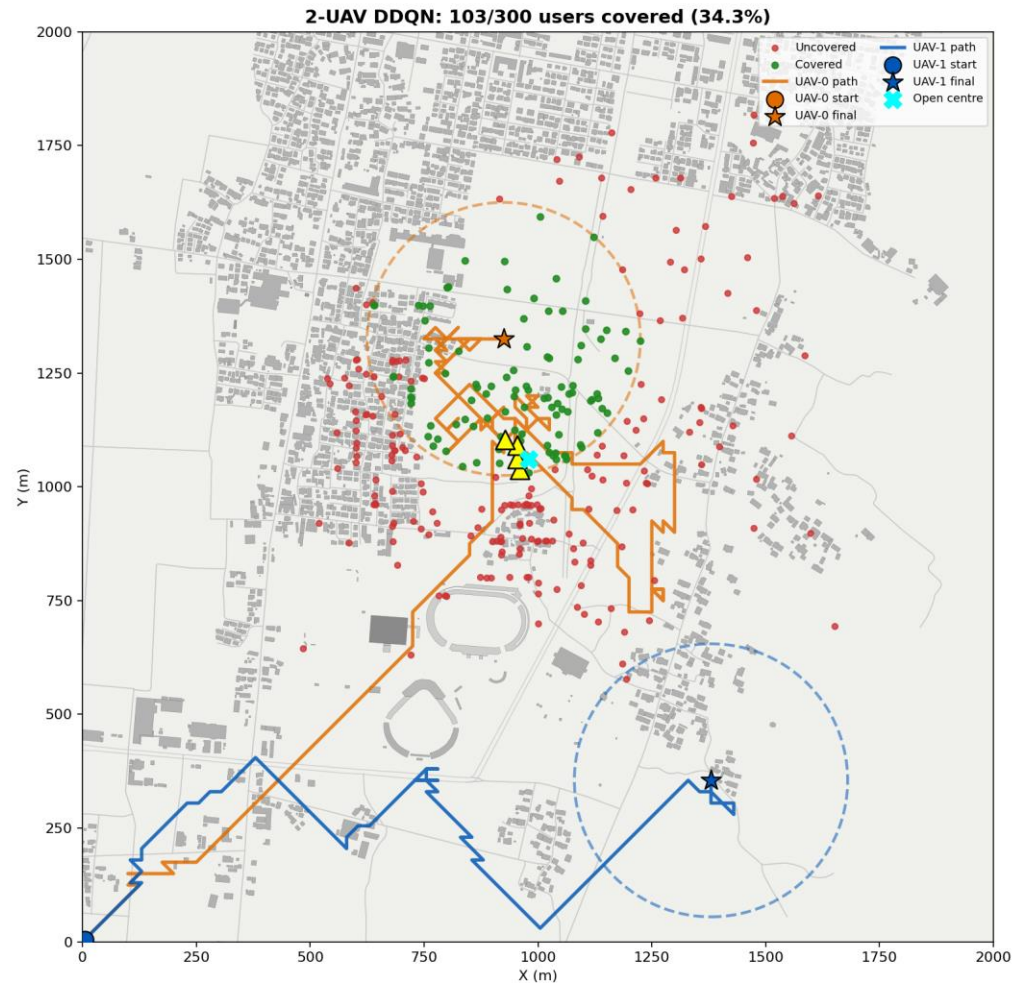
Aizu Athletic Park Stadium
Aizuwakamatsu, Fukushima, Japan

Environment Properties

- Real roads and buildings
- Urban obstacles
- Realistic simulation area

Map Size

2 km × 2 km urban environment



System Model

UAV Model

- Multi UAV
- Height: 120 meters
- Coverage Radius: 400 meters
- UAV Speed: 25 m/s

User Model

- 200 mobile users
- Dynamic movement
- Gauss Markov Mobility Model
- Hotspot-based crowd behaviour

Building Model

- Real building footprints from OpenStreetMap
- Estimated building heights

Dataset Format

OpenStreetMap



Building Shapes



Occupancy Grid



Numerical Arrays



DDQN Agent



UAV Action

Blockage & Communication Model

Model Used

- 3GPP-inspired Urban Macro Model
- Probabilistic LoS/NLoS Model

Why Used?

- Suitable for urban UAV communication
- Models signal blockage by buildings
- Computationally efficient for RL

Communication Factors

- Line-of-Sight (LoS)
- Non-Line-of-Sight (NLoS)
- Path loss
- Urban attenuation

Limitations

- Simplified radio propagation
- Approximate building heights
- No full ray tracing



Fig: UAV LoS/NLoS communication.

User Mobility Model

Mobility Used

- Gauss-Markov Mobility Model
- Hotspot-Based Movement

Why?

- Creates realistic human movement
- Smooth movement behaviour
- Dynamic user clustering

Hotspot Behaviour

- Users gather near stadium/open areas
- Simulates crowd behaviour
- UAV learns to find dense user regions

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Deep Reinforcement Learning

RL Agent

Double Deep Q-Network (Double DQN)

Why Double DQN?

- Reduces Q-value overestimation
- More stable learning
- Better UAV movement decisions

RL Goal

- Maximize long-term coverage reward
- Learn optimal UAV placement automatically

RL Environment Design

Observation Space

- UAV position
- User positions
- Coverage information
- LoS probability
- Building/blockage information

Action Space

- Move Up
- Move Down
- Move Left
- Move Right
- Move Up Right
- Move Up Left
- Move Down Right
- Move Down Left
- Stay

Reward Function

☐ Positive Reward

- Higher user coverage
- Better LoS
- Efficient exploration

☐ Negative Reward

- Low coverage
- Repeated movement
- Poor positioning

Double DQN Architecture

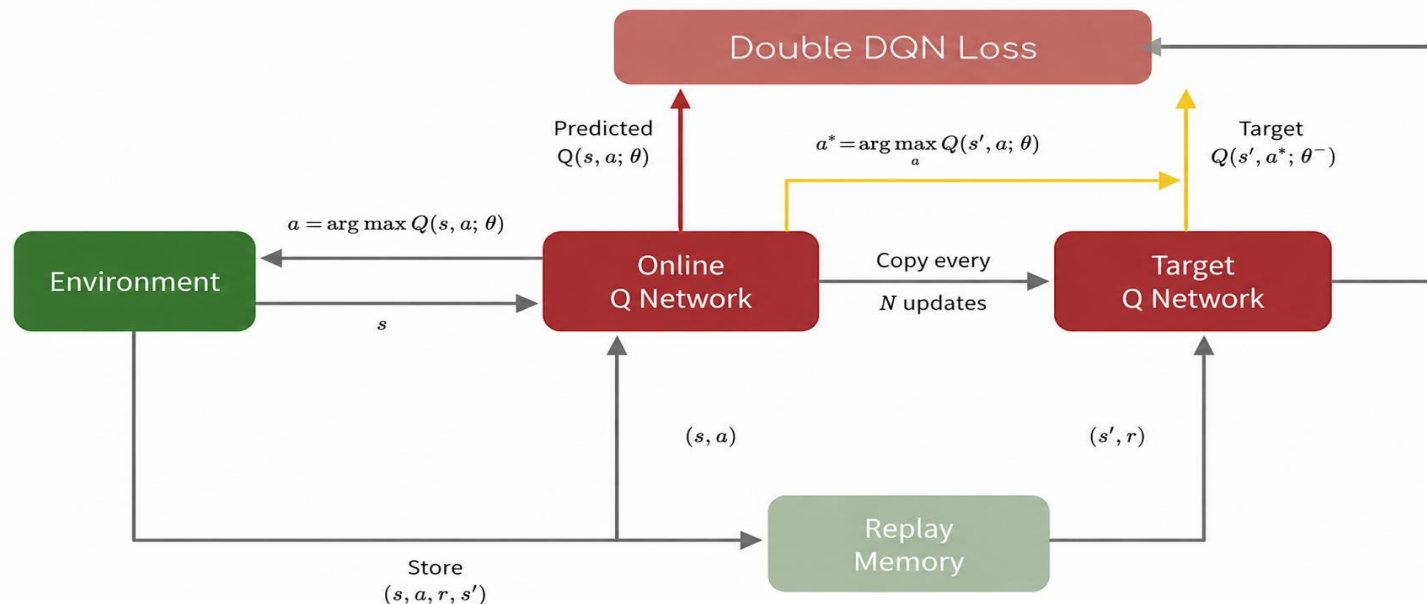


Fig: Double DQN Architecture.

Working Principle

- Main Network selects action
- Target Network evaluates action

Advantage

- Stable learning
- Reduced overestimation
- Better policy learning

Training Process

- Observe state
- Select action
- Receive reward
- Store experience
- Update network

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Simulation results

- Number of UAVs: 2
- Environment: Real city map (OpenStreetMap)
- Users: Randomly distributed
- RL Algorithm: Double DQN
- Episodes: 300
- Coverage Radius: 300 m

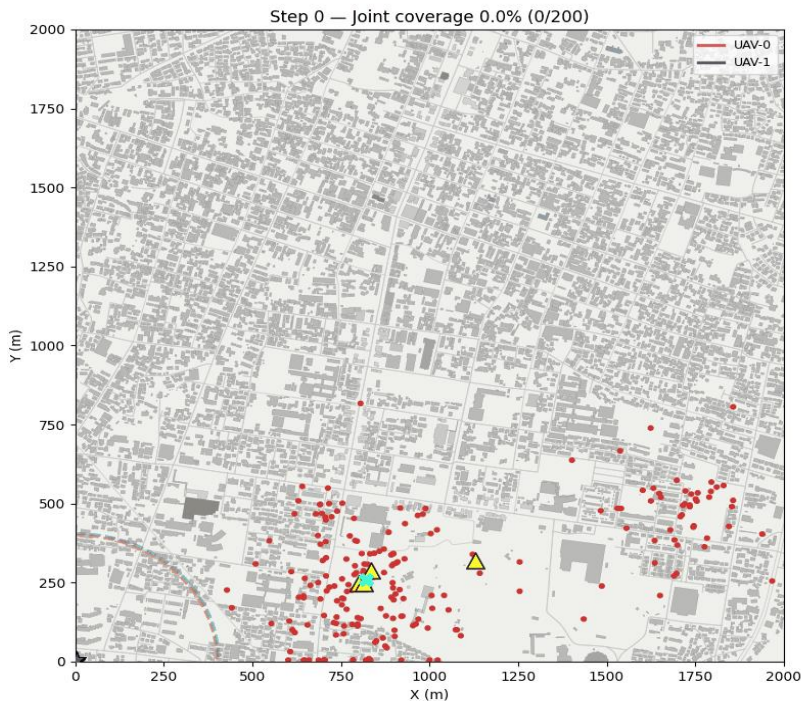


Fig-1: 2D visualization

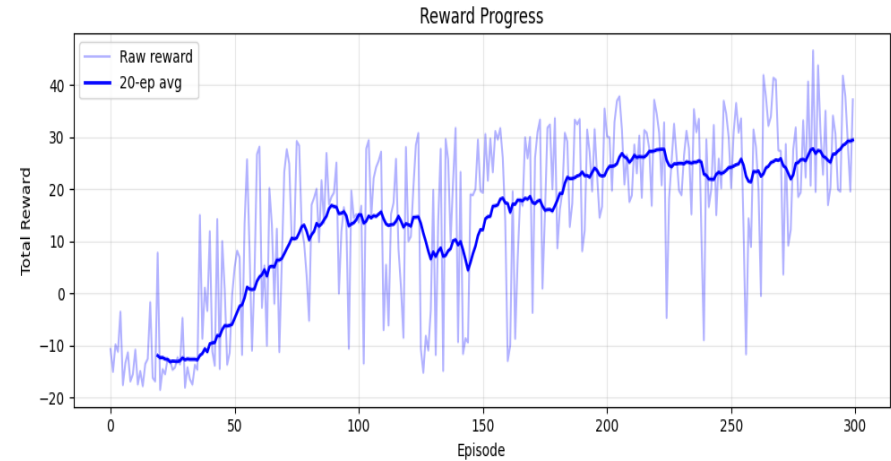


Fig-2: Reward Chart

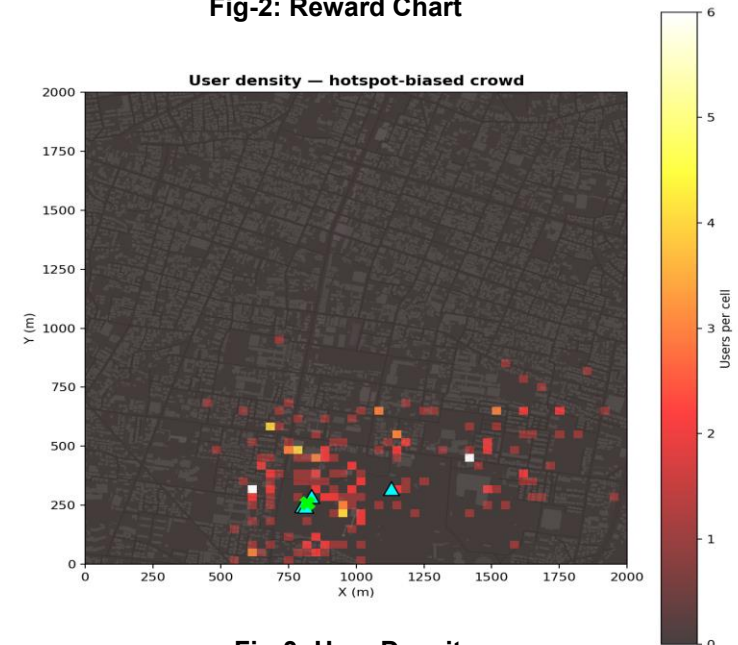


Fig-3: User Density

Discussion & Limitations

Achievements

- Real city map integration
- Intelligent UAV exploration
- Dynamic user mobility
- Urban blockage simulation

Current Limitations

- Simplified propagation model
- Building heights are not explicitly modelled.
- Training instability in some episodes
- Cannot perform true 3D blockage analysis.

Future Work

- PPO implementation
- More advanced propagation models
- Continuous UAV control
- Use OSM building heights or 3D city models.
- Large-scale multi-UAV coordination.

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Conclusion

Summary

- Developed intelligent UAV placement system
- Used Double DQN for autonomous learning
- Simulated realistic urban environment
- Integrated real OpenStreetMap city data
- Considered user mobility and urban blockage

Outcome

- UAV learns coverage optimization autonomously
- Reinforcement Learning improves decision making
- Realistic framework for future UAV communication research

Thank you!